

NT-C-020 | Business Analytics | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	Data & Business
Target learners	Graduates and professionals seeking data-driven business decision roles.
Prerequisites	Basic spreadsheet and business understanding.
Approved duration	14-18 weeks / 160-220 guided hours.
Final outcome	Translate business questions into KPIs, analyses, dashboards and actionable recommendations.
Assessment profile	Data/AI

Approved tools / platforms

Excel, SQL, Power BI, statistics tools and optional Python/pandas.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Require data provenance, cleaning decisions, train/test separation, suitable metrics, reproducibility and responsible interpretation.	Reject data leakage, invented metrics, unsupported causal claims and hidden manual corrections.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-020 | Business Analytics | NT-C-020-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	Objectives, stakeholders, KPIs and hypotheses; Functions, pivots and scenario analysis
Completion evidence	Analytics brief; Analysis workbook
Section weight	20% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 1 - Business framing

Objective: Objectives, stakeholders, KPIs and hypotheses

Observable evidence: Analytics brief

Concept	Approved knowledge statement	Application behaviour	Evidence
Objectives	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when objectives is appropriate; apply it in a Business Analytics task; verify the result.	Analytics brief
stakeholders	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when stakeholders is appropriate; apply it in a Business Analytics task; verify the result.	Analytics brief
KPIs	A defined metric tied to a business objective and decision. In Business Analytics, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when kpis is appropriate; apply it in a Business Analytics task; verify the result.	Analytics brief
hypotheses	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when hypotheses is appropriate; apply it in a Business Analytics task; verify the result.	Analytics brief

Module 2 - Excel analytics

Objective: Functions, pivots and scenario analysis

Observable evidence: Analysis workbook

Concept	Approved knowledge statement	Application behaviour	Evidence
Functions	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when functions is appropriate; apply it in a Business Analytics task; verify the result.	Analysis workbook
pivots	An interactive aggregation used to summarise and explore tabular data. In Business Analytics, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when pivots is appropriate; apply it in a Business Analytics task; verify the result.	Analysis workbook
scenario analysis	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when scenario analysis is appropriate; apply it in a Business Analytics task; verify the result.	Analysis workbook

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Analytics brief; Analysis workbook, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-020 | Business Analytics | NT-C-020-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	Business queries, joins and aggregation; Variation, correlation, forecasting basics
Completion evidence	SQL case; Insight note
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 3 - SQL

Objective: Business queries, joins and aggregation

Observable evidence: SQL case

Concept	Approved knowledge statement	Application behaviour	Evidence
Business queries	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when business queries is appropriate; apply it in a Business Analytics task; verify the result.	SQL case
joins	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when joins is appropriate; apply it in a Business Analytics task; verify the result.	SQL case
aggregation	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when aggregation is appropriate; apply it in a Business Analytics task; verify the result.	SQL case

Module 4 - Statistics

Objective: Variation, correlation, forecasting basics

Observable evidence: Insight note

Concept	Approved knowledge statement	Application behaviour	Evidence
Variation	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when variation is appropriate; apply it in a Business Analytics task; verify the result.	Insight note
correlation	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when correlation is appropriate; apply it in a Business Analytics task; verify the result.	Insight note
forecasting basics	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when forecasting basics is appropriate; apply it in a Business Analytics task; verify the result.	Insight note

Remediation: If the learner cannot explain the purpose, complete the stated task or provide SQL case; Insight note, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-020 | Business Analytics | NT-C-020-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	Quality, transformation and reproducibility; Modelling, DAX and dashboard design
Completion evidence	Clean dataset; Interactive dashboard
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 5 - Data preparation

Objective: Quality, transformation and reproducibility

Observable evidence: Clean dataset

Concept	Approved knowledge statement	Application behaviour	Evidence
Quality	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when quality is appropriate; apply it in a Business Analytics task; verify the result.	Clean dataset
transformation	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when transformation is appropriate; apply it in a Business Analytics task; verify the result.	Clean dataset
reproducibility	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when reproducibility is appropriate; apply it in a Business Analytics task; verify the result.	Clean dataset

Module 6 - Power BI

Objective: Modelling, DAX and dashboard design

Observable evidence: Interactive dashboard

Concept	Approved knowledge statement	Application behaviour	Evidence
Modelling	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when modelling is appropriate; apply it in a Business Analytics task; verify the result.	Interactive dashboard
DAX	The expression language used to define measures and calculations in power bi. In Business Analytics, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when dax is appropriate; apply it in a Business Analytics task; verify the result.	Interactive dashboard
dashboard design	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when dashboard design is appropriate; apply it in a Business Analytics task; verify the result.	Interactive dashboard

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Clean dataset; Interactive dashboard, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-020 | Business Analytics | NT-C-020-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	Segmentation, trends and what-if analysis; Executive presentation and implementation plan
Completion evidence	Recommendation memo; Business analytics case
Section weight	30% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 7 - Decision analysis

Objective: Segmentation, trends and what-if analysis

Observable evidence: Recommendation memo

Concept	Approved knowledge statement	Application behaviour	Evidence
Segmentation	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when segmentation is appropriate; apply it in a Business Analytics task; verify the result.	Recommendation memo
trends	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when trends is appropriate; apply it in a Business Analytics task; verify the result.	Recommendation memo
what-if analysis	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when what-if analysis is appropriate; apply it in a Business Analytics task; verify the result.	Recommendation memo

Module 8 - Capstone

Objective: Executive presentation and implementation plan

Observable evidence: Business analytics case

Concept	Approved knowledge statement	Application behaviour	Evidence
Executive presentation	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when executive presentation is appropriate; apply it in a Business Analytics task; verify the result.	Business analytics case
implementation plan	A defined competency within Business Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when implementation plan is appropriate; apply it in a Business Analytics task; verify the result.	Business analytics case

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Recommendation memo; Business analytics case, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-020 | Business Analytics | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-020-SK01	Business framing	Prerequisite foundation	Analytics brief	Critical
NT-C-020-SK02	Excel analytics	M01 plus earlier foundations	Analysis workbook	Critical
NT-C-020-SK03	SQL	M02 plus earlier foundations	SQL case	Supporting
NT-C-020-SK04	Statistics	M03 plus earlier foundations	Insight note	Supporting
NT-C-020-SK05	Data preparation	M04 plus earlier foundations	Clean dataset	Supporting
NT-C-020-SK06	Power BI	M05 plus earlier foundations	Interactive dashboard	Critical
NT-C-020-SK07	Decision analysis	M06 plus earlier foundations	Recommendation memo	Supporting
NT-C-020-SK08	Capstone	M07 plus earlier foundations	Business analytics case	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-020 | Business Analytics | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-020-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of Business framing and Excel analytics?	The learner should provide Analytics brief; Analysis workbook and explain how the work applies Objectives, stakeholders, KPIs and hypotheses; Functions, pivots and scenario analysis. Recall alone is insufficient.
NT-C-020-S01-EX02	Scenario/remediation	A learner reports completing Excel analytics but cannot produce Analysis workbook. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-020-S01, request Analysis workbook, and allow a new equivalent test only after remediation.
NT-C-020-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of SQL and Statistics?	The learner should provide SQL case; Insight note and explain how the work applies Business queries, joins and aggregation; Variation, correlation, forecasting basics. Recall alone is insufficient.
NT-C-020-S02-EX02	Scenario/remediation	A learner reports completing Statistics but cannot produce Insight note. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-020-S02, request Insight note, and allow a new equivalent test only after remediation.
NT-C-020-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of Data preparation and Power BI?	The learner should provide Clean dataset; Interactive dashboard and explain how the work applies Quality, transformation and reproducibility; Modelling, DAX and dashboard design. Recall alone is insufficient.
NT-C-020-S03-EX02	Scenario/remediation	A learner reports completing Power BI but cannot produce Interactive dashboard. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-020-S03, request Interactive dashboard, and allow a new equivalent test only after remediation.
NT-C-020-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Decision analysis and Capstone?	The learner should provide Recommendation memo; Business analytics case and explain how the work applies Segmentation, trends and what-if analysis; Executive presentation and implementation plan. Recall alone is insufficient.
NT-C-020-S04-EX02	Scenario/remediation	A learner reports completing Capstone but cannot produce Business analytics case. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-020-S04, request Business analytics case, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Data/AI blueprint. Human-review every scored item before initial production release.

NT-C-020 | Business Analytics | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	Sales KPI review	Excel and business metrics	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	Customer performance dashboard	SQL and Power BI	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	Branch growth analysis	End-to-end analytics and recommendations	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Require data provenance, cleaning decisions, train/test separation, suitable metrics, reproducibility and responsible interpretation.

NT-C-020 | Business Analytics | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	Junior Business Analyst - Analytics	KPIs 34% [critical]; Excel 33% [critical]; SQL 33%	Business case	If a critical skill is below 60 or business case is missing, return Improve First or Needs Evidence with the linked module/project.
2	Business Intelligence Analyst - Junior	Modelling 34% [critical]; DAX 33% [critical]; dashboards 33%	Published report	If a critical skill is below 60 or published report is missing, return Improve First or Needs Evidence with the linked module/project.
3	Operations Analyst	Trends 34% [critical]; scenarios 33% [critical]; recommendations 33%	Decision memo	If a critical skill is below 60 or decision memo is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-020 | Business Analytics | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is Business Analytics for?	Graduates and professionals seeking data-driven business decision roles.
What should I know before starting?	Basic spreadsheet and business understanding.
How long is the approved learning journey?	14-18 weeks / 160-220 guided hours.
What will I be able to demonstrate?	Translate business questions into KPIs, analyses, dashboards and actionable recommendations.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
Business Analytics knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

```
course_code=NT-C-020; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12
```